Ex 4

Program :

```
import numpy as np


# Sigmoid function
def sigmoid(x):
  return 1 / (1 + np.exp(-x))


def sigmoid_derivative(x):
  return x * (1 - x)


# XOR dataset
X = np.array([[0, 0],
       [0, 1],
       [1, 0],
       [1, 1]])


y = np.array([[0], [1], [1], [0]])


np.random.seed(1)


# Initialize weights
w1 = np.random.uniform(size=(2, 4))
w2 = np.random.uniform(size=(4, 1))


learning_rate = 0.5


# Training using Backpropagation
```

```
for epoch in range(10000):
    # Forward propagation
    hidden = sigmoid(np.dot(X, w1))
    output = sigmoid(np.dot(hidden, w2))

    # Error
    error = y - output

    # Backpropagation
    d_output = error * sigmoid_derivative(output)
    d_hidden = d_output.dot(w2.T) * sigmoid_derivative(hidden)

    # Update weights
    w2 += hidden.T.dot(d_output) * learning_rate
    w1 += X.T.dot(d_hidden) * learning_rate

print("Actual Output:")
print(y)

print("\nPredicted Output:")
print(np.round(output))
```

output :

```
•••  Actual Output:
     [[0]
      [1]
      [1]
      [0]]

     Predicted Output:
     [[0.]
      [1.]
      [1.]
      [0.]]
```